2. Two six-sided dice are thrown sequentially, and the face values that come up are recorded.

30' **a.** List the sample space.

30' **b.** List the elements that make up the following events: (1) A = the sum of the two values is at least 5, (2) B = the value of the first die is higher than the value of the second, (3) C = the first value is 4.

30' **c.** List the elements of the following events: (1) $A \cap C$, (2) $B \cup C$, (3) $A \cap (B \cup C)$.

a. sample space = {(1,1), (1,2), (1,3), (1,4), (1,5), (1,6)
(2,1), (2,2), (2,3), (2,4), (2,5), (2,6)
(3,1), (3,2), (3,3), (3,4), (3,5), (3,6)
(4,1), (4,2), (4,3), (4,4), (4,5), (4,6)
(5,1), (5,2), (5,3), (5,4), (5,5), (5,6)
(6,1), (6,2), (6,3), (6,4), (6,5), (6,6)}

b. A = {(1,4), (1,5), (1,6)
(2,3), (2,4), (2,5), (2,6)
(3,2), (3,3), (3,4), (3,5), (3,6)
(4,1), (4,2), (4,3), (4,4), (4,5), (4,6)
(5,1), (5,2), (5,3), (5,4), (5,5), (5,6)
(6,1), (6,2), (6,3), (6,4), (6,5), (6,6)}

B = {(2,1), (3,1), (3,2), (4,1), (4,2), (4,3), (5,1), (5,2), (5,3), (5,4)
(6,1), (6,2), (6,3), (6,4), (6,5)}

C = {(4,1), (4,2), (4,3), (4,4), (4,5), (4,6)}

∴ (1) $A \cap C = C$ = {(4,1), (4,2) (4,3), (4,4), (4,5), (4,6)}.

(2) $B \cup C$ = {(2,1), (3,1), (3,2), (4,1), (4,2)
(4,3), (4,4), (4,5), (4,6), (5,1)
(5,2), (5,3), (5,4), (6,1), (6,2)
(6,3), (6,4), (6,5)}.

(3) $A \cap (B \cup C)$ = {(3,2), (1,1), (4,2), (4,3), (4,4), (4,5), (4,6), (5,1)
(5,2), (5,3), (5,4), (6,1), (6,2), (6,3), (6,4), (6,5)}

5. Let A and B be arbitrary events. Let C be the event that either A occurs or B occurs, but not both. Express C in terms of A and B using any of the basic operations of union, intersection, and complement.

10'

$$C = (A \cup B) \cap (\overline{A \cap B})$$

或者 $(A^c \cap B) \cup (B^c \cap A)$

或者 $((A \cup B) \cap A^c) \cup ((A \cup B) \cap B^c)$

P27: Q7 , Q8

8'

7. Prove Bonferroni's inequality:

$$P(A \cap B) \geq P(A) + P(B) - 1$$

10'

8. Prove that

$$P\left(\bigcup_{i=1}^{n} A_i\right) \leq \sum_{i=1}^{n} P(A_i)$$

7. Proof: $1 - P(A \cap B) = P(\overline{A \cap B})$

$= P(\bar{A} \cup \bar{B}) \leq P(\bar{A}) + P(\bar{B})$

$= 1 - P(A) + 1 - P(\bar{B})$

$\Rightarrow P(A \cap B) \geq P(A) + P(B) - 1$

8. Proof: Let $A_0 = \phi$. $B_1 = A_1$, $B_2 = A_2 - B_1$, $B_3 = A_3 - B_1 - B_2$, $\cdots$,

$B_n = A_n - \bigcup_{i=1}^{n-1} B_i$

Then. $B_i \subset A_i$, and $\{B_n\}$ are pairwise disjoint,

and $\bigcup_{i=1}^{n} A_i = \bigcup_{i=1}^{n} B_i$

We have $\underline{P(\bigcup_{i=1}^{n} A_i)} = P(\bigcup_{i=1}^{n} B_i) = \sum_{i=1}^{n} P(B_i) \underline{\leq \sum_{i=1}^{n} P(A_i)}$

Hence. $P(\bigcup_{i=1}^{n} A_i) \leq \sum_{i=1}^{n} P(A_i)$

Supplementary questions:

7' 1. For three random events A, B, C, $P(A) = P(B) = P(C) = \frac{1}{4}$, $P(AB) = P(BC) = 0$, $P(AC) = \frac{1}{8}$. What is the probability of at least one of events A, B, C happens?

$P(A \cup B \cup C)$

Solution:

$\because P(AB) = 0$. $ABC \subset AB$. $\Rightarrow P(ABC) = 0$

$$P(A \cup B \cup C) = P(A) + P(B) + P(C) - P(AB) - P(BC) - P(AC) + P(ABC)$$
$$= \frac{1}{4} + \frac{1}{4} + \frac{1}{4} - 0 - 0 - \frac{1}{8} + 0$$
$$= \frac{5}{8}$$

10' 2. Let A, B be two events that satisfy $P(AB) = P(\overline{A}\,\overline{B})$, and $P(A) = p$. Find $P(B)$.

Solution: We have known $P(AB) = P(\bar{A}\bar{B})$, $P(A) = P$.

$$1 - P(A \cap B) = P(\overline{A \cap B})$$
$$= P(\bar{A} \cup \bar{B})$$
$$= P(\bar{A}) + P(\bar{B}) - P(\bar{A}\bar{B})$$
$$= 1 - P(A) + 1 - P(B) - P(AB)$$

Hence. $1 - P(AB) = 2 - P - P(AB) - P(B)$

$\Rightarrow P(B) = 1 - P$.